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TECHNICAL REPORT 04

EnergieBee core Planning design compatibility

What the core already covers and what is left to build

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Content is reproduced from the source document `docs/planning-design-compatibility.md`, which remains the editable original.

REPORT OVERVIEW

How the core fits the "Predbat in Node.js + SQL" planning design and the comparison document that framed the two systems as a ledger of the past and a controller of the future. The core is the ledger. The planner and executor are not built.

Verdict

Part of the planning design	In the core	Status
Time grid: 5-minute steps, slots keyed by UTC start	Same, with a per-site `slot_s`	Compatible
Input tables replaced by the ledger (rates, PV forecast, load history, telemetry)	Covered by `rate_interval`, `forecast_fact`, `site_hh`, `reading`	Compatible
Versioned facts with provenance	Append-only facts with a source, a learned-at time and a raw payload	Compatible (better than the design)
Data quality gate	Slot states and quality flags	Partial: no gate yet
`site` with timezone and plan interval	`site` with timezone, market, currency and `slot_s`	Compatible
Device registry (inverter, battery, meters)	`device` plus ranked `site_attribute`	Partial
`site_setting` tunables (keep level, solar weight)	Not present	Missing
`tenant`, `credential`	Not present	Missing
`plan_run`, `plan_slot`, `current_plan`	Not present	Missing
`inverter_command` and executor with read-back	Not present	Missing
`event_log`	`sync_run` and the revision log (not the same)	Partial
Scheduler with SKIP LOCKED and worker pool	Single serial sync	Missing
Edge agent, MQTT topics, hybrid mode	Not present	Missing
Planner as a pure function	Not built	Missing

Example 1: replacing the design's input tables

The design's `tariff_rate` copies a price per site per slot. The core stores each price once and joins it.

Design table	Replaced by	Why
`tariff_rate` (site, direction, slot, p/kWh)	`rate_interval` + `tariff_agreement`	Prices are shared by every home on a tariff
`pv_forecast` (one row per slot)	`forecast_fact`	Every forecast issue is kept and scored
`load_history` (5-minute buckets)	`reading` and `site_hh`	Already stored and checked
`telemetry` (wide row per inverter)	`reading` (narrow, one metric per row)	A new metric needs no migration

Example 2: a slot, side by side

The planner's `plan_slot` and the core's ledger row share one key. A plan for the slot starting 2026-10-24T00:00:00Z would line up like this (illustrative values):

Key	`plan_slot` (planned)	`site_hh` (what happened)
`slot_start`	1792800000	1792800000
Action	charge, limit 100%	n/a
Import	1.20 kWh predicted	measured from the meter
Cost	predicted	priced from the tariff in force

The design proposes storing each plan's predictions as a forecast source so the existing accuracy scoring grades the planner. The core's forecast scoring already does this for its own forecasts.

Example 3: a command

The design's `inverter_command` carries a desired state, a status and a read-back link. In the core's terms it would be one more kind of fact:

Field	Example
Device	the home's inverter
Desired	charge window 23:30 to 05:30, target 100%
Valid until	05:30 (a late command is dropped, never applied stale)
Status	pending, sent, acked or failed
Read-back	the schedule read from the inverter, kept as a raw payload

Nothing here exists yet. The core already has the raw-payload store and the idea of a checked read.

What the core adds that the design lacks

- Provenance on every figure, and labels for sourced, provider, user, placeholder and synthetic data.
- Prices shared per tariff, not copied per home.
- Bitemporal "as known at" rebuilds, so a plan can be reproduced from the facts it used.
- Country, currency and region handling, and tariff shapes the design never covered (French off-peak, Tempo, fixed charges).

What the design adds that the core lacks

- A control path to the device.
- Multi-tenancy, a scheduler and a worker pool.
- An edge agent, MQTT topics and store-and-forward.
- Tunable settings per site.

Suggested order to close the gaps

1. `plan_run` and `plan_slot`, written only on change.
2. The planner as a pure function, run in shadow mode and scored.
3. `site_setting` for the planner's tunables.
4. The command path with expiry and read-back.
5. Tenant scoping and the scheduler when more than one home needs it.
6. The edge agent last.

See `docs/capabilities-and-roadmap.md` for effort and `docs/predbat-slot-compatibility.md` for the slot detail.